Blood 144 (2024) 677–679



The 66th ASH Annual Meeting Abstracts

ORAL ABSTRACTS

654.MULTIPLE MYELOMA: PHARMACOLOGIC THERAPIES

Mezigdomide (MEZI) in Novel-Novel Combinations for Relapsed or Refractory Multiple Myeloma (RRMM): Preliminary Results from the CA057-003 Trial

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Introduction: MEZI, a novel, potent oral CELMoD™ agent, induces rapid and maximal degradation of Ikaros and Aiolos compared with immunomodulatory drugs (IMiDs®), resulting in direct tumoricidal and immunomodulatory effects in myeloma cells. The CA057-003 phase 1/2 trial (NCT05372354) is evaluating all-oral, novel-novel targeted triplet combination regimens using a MEZI plus dexamethasone (DEX) (MEZId) backbone in patients (pts) with RRMM. The third agent in each combination intervenes on a key oncogenic pathway identified by The Myeloma Genome Project to be upregulated in RRMM: the EZH2 inhibitor tazemetostat (TAZ) for PRC2 complex dysregulation, the BET inhibitor BMS-986158 for CKS1b (located on Chr 1q) amplification, and the MEK inhibitor trametinib (TRAM) for RAS-RAF-MEK-ERK activation. Here we report preliminary results from the CA057-003 dose-finding cohorts of MEZId combined with TAZ, BMS-986158, or TRAM in pts with RRMM.
Methods: Eligible pts had RRMM with documented progressive disease (PD) during or after the last regimen, ECOG performance status score ≤1, and were intolerant to or ineligible for all available established therapies. Oral MEZI was given daily (QD) at escalating doses of 0.3, 0.6, or 1.0 mg on days (D)1-21 of each 28-D cycle with 40 mg (20 mg if ≥75 y of age) weekly DEX plus 800 mg twice daily oral TAZ on D1-28; 2 or 3 mg QD oral BMS-986158 5D-on-2D-off D1-14; or 1.5 or 2 mg QD oral TRAM on D1-21. Primary objectives included defining the recommended phase 2 dose and dosing schedule, and evaluating safety. Secondary objectives included efficacy and pharmacokinetics. There was no biomarker screening.

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Results: As of May 8, 2024, 14 pts received MEZId + TAZ, 16 MEZId + BMS-986158, and 15 MEZId + TRAM. Across all cohorts, median (range) age was 63 (37-83) y and median time since initial diagnosis was 7.5 (2.0-18.4) y. Eight (18%) pts were Black/African American, 35 (78%) White, and race was not collected/unknown (NA) in 2 (4%); 6 (13%) were Hispanic/Latino and 36 (80%) were not (NA = 3 [7%] pts); 56% were in the United States. Extramedullary plasmacytomas were present in 16 (36%) pts. The median number of prior antimyeloma regimens was 5 (2-20). Prior therapies included autologous stem cell transplantation (78%), IMiD agents (100%), proteasome inhibitors (PIs; 100%), anti-CD38 monoclonal antibodies (mAbs; 100%), and T cell-redirecting therapy (58%); 82% had triple-class refractory disease (to an IMiD agent, PI, and anti-CD38 mAb).
At data cutoff, 7 (50%) pts continued on treatment in the MEZId + TAZ cohort, 8 (50%) in the MEZId + BMS-986158 cohort, and 10 (67%) in the MEZId + TRAM cohort. The main reason for discontinuation in all 3 cohorts was PD. Median follow-up was 4.1 (1.0-11.0) mo (MEZId + TAZ), 2.9 (1.0-6.5) mo (MEZId + BMS-986158), and 3.6 (0-10.6) mo (MEZId + TRAM).
The most frequent grade (Gr) 3/4 treatment-emergent adverse event (TEAE) across all 3 cohorts was neutropenia (43-73%); Gr 3/4 non-hematologic TEAEs were low or absent. Three pts had dose-limiting toxicities (1 with 0.3 mg MEZId + BMS-986158, 1 with 1.0 mg MEZId + BMS-986158, and 1 with 1.0 mg MEZId + TRAM).
In the efficacy-evaluable population, overall response rate (≥ partial response [PR]) was 54% (7/13 pts) with MEZId + TAZ; 36% (5/14) with MEZId + BMS-986158; and 92% (11/12) with MEZId + TRAM. In the MEZId + TAZ and MEZId + BMS-986158 cohorts, deeper responses (≥ very good PR), including 1 stringent complete response (sCR) with MEZId + TAZ, were observed with 1.0 mg MEZI, and in the MEZId + TRAM cohort with ≥0.6 mg MEZI (including 1 sCR), and 100% (5/5 pts) at 1.0 mg MEZI were ongoing responses with this combination.
Exposures increased in a dose-linear manner over the dose range and were consistent across treatment cohorts, demonstrating no drug-drug interaction between MEZI and novel therapeutic agents. MEZI remained pharmacodynamically active, inducing Ikaros/Aiolos degradation and B-cell reduction with all combination agents at all dose levels, with the greatest effect observed at MEZI 1.0 mg.
Conclusions: MEZId combined with the novel therapeutic agents TAZ, BMS-986158, or TRAM showed promising efficacy and a manageable safety profile in patients with RRMM. No new safety signals were identified, with neutropenia being the most common Gr 3/4 TEAE in all 3 cohorts. These results provide a rationale for further exploration of these novel all-oral combinations. Accrual continues and updated results will be presented at the meeting.

Disclosures Costa: *Janssen:* Consultancy, Honoraria, Research Funding; *Abbvie:* Consultancy, Honoraria, Research Funding; *Amgen:* Consultancy, Honoraria, Research Funding; *BMS:* Consultancy, Honoraria, Research Funding; *Caribou:* Research Funding; *Genentech, Inc.:* Consultancy, Honoraria, Research Funding; *Adaptive biotechnoligies:* Honoraria; *Pfizer:* Consultancy, Honoraria; *Sanofi:* Consultancy, Honoraria. **Schjesvold:** *Takeda:* Consultancy, Other: Honoraria for lectures and educational material; *Amgen:* Other: Honoraria for lectures and educational material; *Targovax:* Research Funding; *Celgene:* Consultancy, Other: Honoraria for lectures and educational material, Research Funding; *AbbVie:* Consultancy, Other: Honoraria for lectures and educational material; *GlaxoSmithKline:* Consultancy, Honoraria, Research Funding; *Bristol Myers Squibb:* Consultancy, Other: Honoraria for lectures and educational material; *GSK:* Consultancy, Honoraria, Research Funding; *Sanofi:* Consultancy, Other: Honoraria for lectures and educational material, Research Funding; *SkylineDx:* Other: Honoraria for lectures and educational material; *Novartis:* Other: Honoraria for lectures and educational material; *Daiichi Sankyo:* Other: Honoraria for lectures and educational material; *Janssen-Cilag:* Consultancy, Other: Honoraria for lectures and educational material, Research Funding; *Oncopeptides:* Consultancy, Other: Honoraria for lectures and educational material, Research Funding; *Pfizer:* Other: Honoraria for lectures and educational material; *Schain:* Other: Honoraria for lectures and educational material; *Skylite:* Other: Honoraria for lectures and educational material. **Popat:** *BMS:* Honoraria; *Abbvie:* Honoraria; *Janssen:* Honoraria, Speakers Bureau; *Sanofi:* Honoraria; *GSK:* Honoraria, Research Funding; *Pfizer:* Honoraria, Research Funding, Speakers Bureau. **Usmani:** *Bristol-Myers Squibb:* Consultancy, Research Funding; *Array Biopharma:* Research Funding; *Gracell:* Consultancy; *Abbvie:* Consultancy, Research Funding; *SkylineDX:* Consultancy, Research Funding; *Amgen:* Consultancy, Research Funding; *Pharmacyclics:* Research Funding; *SeaGen:* Consultancy, Research Funding; *Sanofi:* Consultancy, Research Funding; *SecuraBio:* Consultancy; *Gilead:* Research Funding; *Oncopeptides:* Consultancy; *Pfizer:* Consultancy; *Sanofi:* Consultancy, Research Funding; *Bristol-Myers Squibb - Celgene::* Consultancy, Research Funding; *EdoPharma:* Consultancy; *Takeda:* Consultancy, Research Funding; *TeneoBio:* Consultancy; *GSK:* Consultancy, Research Funding; *Merck:* Research Funding; *Genentech:* Consultancy; *Bristol-Myers Squibb - Celgene:* Consultancy, Research Funding; *Johnson & Johnson - Janssen:* Consultancy, Research Funding. **Ali:** *Bristol Myers Squibb:* Consultancy, Research Funding; *Pfizer/IMF:* Research Funding; *Sanofi:* Consultancy, Membership on an entity's Board of Directors or advisory committees. **Chu:** *Pfizer:* Honoraria; *Janssen:* Honoraria; *AstraZeneca:* Honoraria; *Amgen:* Honoraria; *Kite/Gilead:* Honoraria; *Sanofi:* Honoraria; *Amgen:* Consultancy; *BMS:* Honoraria. **Hartley-Brown:** *Karyopharm:* Consultancy, Honoraria, Research Funding; *AbbVie:* Consultancy, Honoraria, Research Funding; *GSK:* Consultancy, Honoraria, Research Funding; *Multiple Myeloma Research Foundation:* Consultancy, Honoraria, Research Funding; *Cancer Care:* Consultancy, Honoraria; *Janssen:* Consultancy, Honoraria, Research Funding; *Sanofi:* Consultancy, Honoraria, Research Funding; *Bristol Myers Squibb/Celgene:* Consultancy, Honoraria, Research Funding. **Bahlis:** *AbbVie, Amgen, Bristol Myers Squibb, Celgene, Janssen, Karyopharm Therapeutics, Kyte, Novartis, Pfizer, Roche, Sanofi, Takeda:* Consultancy; *AbbVie, Amgen, Bristol Myers Squibb, Celgene, Janssen, GlaxoSmithKline, Genentech, Karyopharm Therapeutics, Kyte, Novartis, Pfizer, Roche, Sanofi, Takeda:* Membership on an entity's Board of Directors or advisory committees; *Janssen, Pfizer:* Research Funding; *AbbVie, Amgen, Bristol Myers Squibb, Celgene, Janssen, GlaxoSmithKline, Genentech, Karyopharm Therapeutics, Kyte, Novartis, Pfizer, Roche, Sanofi, Takeda:* Honoraria. **Oriol:** *Sanofi:* Honoraria, Speakers Bureau;

Pfizer, Amgen, Oncopeptides: Honoraria; *GSK:* Honoraria, Speakers Bureau; *Bristol Myers Squibb/Celgene:* Honoraria, Speakers Bureau; *Johnson & Johnson, Janssen:* Honoraria, Speakers Bureau. **Lopez:** *BMS, Pfizer, Janssen, Amgen, Roche, Kite:* Consultancy; *BMS, Pfizer, Janssen, Amgen, Roche, Kite:* Honoraria; *Pfizer, Janssen, Incity:* Research Funding. **Ocio:** *Takeda:* Consultancy, Honoraria; *Regeneron:* Honoraria; *Pfizer:* Consultancy, Honoraria; *Bristol Myers Squibb:* Consultancy, Honoraria; *Sanofi:* Consultancy, Honoraria; *Oncopeptides:* Consultancy, Honoraria, Research Funding; *Menarini:* Consultancy, Honoraria; *AstraZeneca:* Consultancy; *Janssen:* Consultancy, Honoraria, Speakers Bureau; *Amgen:* Honoraria; *GSK:* Consultancy, Honoraria, Research Funding; *AbbVie:* Consultancy, Honoraria. **Ramasamy:** *Bristol-Myers Squibb:* Other: Travel; *Oncopeptides:* Honoraria, Membership on an entity's Board of Directors or advisory committees; *Pfizer:* Honoraria, Membership on an entity's Board of Directors or advisory committees; *Recordati:* Honoraria, Membership on an entity's Board of Directors or advisory committees, Other: Travel; *Menarini:* Membership on an entity's Board of Directors or advisory committees; *Stemline:* Membership on an entity's Board of Directors or advisory committees; *AbbVie:* Honoraria, Membership on an entity's Board of Directors or advisory committees; *Takeda:* Honoraria, Membership on an entity's Board of Directors or advisory committees, Other: Travel; *Amgen:* Honoraria, Membership on an entity's Board of Directors or advisory committees, Other: Travel; *Adaptive Biotech:* Honoraria, Membership on an entity's Board of Directors or advisory committees; *Johnson & Johnson - Janssen:* Honoraria, Membership on an entity's Board of Directors or advisory committees; *Bristol-Myers Squibb - Celgene:* Honoraria, Membership on an entity's Board of Directors or advisory committees, Other: Travel, Research Funding; *GSK:* Honoraria, Membership on an entity's Board of Directors or advisory committees, Research Funding; *Sanofi:* Membership on an entity's Board of Directors or advisory committees, Research Funding. **Reece:** *Janssen, BMS, Sanofi, ORIC, Princess Margaret Cancer Centre:* Other: Grants; *BMS, Janssen, Sanofi, GSK, Pfizer:* Consultancy; *BMS, Janssen, Takeda, Pfizer:* Honoraria; *BMS:* Membership on an entity's Board of Directors or advisory committees. **Searle:** *DarkBlue Therapeutics:* Consultancy, Membership on an entity's Board of Directors or advisory committees; *BMS:* Consultancy, Honoraria, Membership on an entity's Board of Directors or advisory committees; *Pfizer:* Honoraria, Speakers Bureau; *Janssen:* Honoraria, Speakers Bureau; *Shattuck Labs:* Consultancy, Membership on an entity's Board of Directors or advisory committees; *Sanofi:* Consultancy, Membership on an entity's Board of Directors or advisory committees; *Incyte:* Research Funding; *Jazz:* Speakers Bureau; *AbbVie:* Honoraria, Speakers Bureau; *Beigene:* Honoraria; *Nurix:* Honoraria. **Gaudy:** *BMS:* Current Employment, Current equity holder in publicly-traded company. **Kurtova:** *Bristol Myers Squibb:* Current Employment, Current equity holder in publicly-traded company. **Zhang:** *Bristol Myers Squibb:* Current Employment, Current equity holder in publicly-traded company. **Sarmiento:** *Bristol Myers Squibb:* Current Employment, Current equity holder in publicly-traded company. **Dietrich:** *Bristol Myers Squibb:* Current Employment, Current equity holder in publicly-traded company. **Katz:** *Bristol Myers Squibb:* Current Employment, Current equity holder in publicly-traded company. **Pourdehnad:** *Bristol Myers Squibb:* Current Employment, Current equity holder in publicly-traded company. **Richardson:** *Oncopeptides:* Research Funding; *Celgene/Bristol Myers Squibb, GSK, Karyopharm Therapeutics, Oncopeptides, Regeneron, Sanofi:* Consultancy.

https://doi.org/10.1182/blood-2024-198403